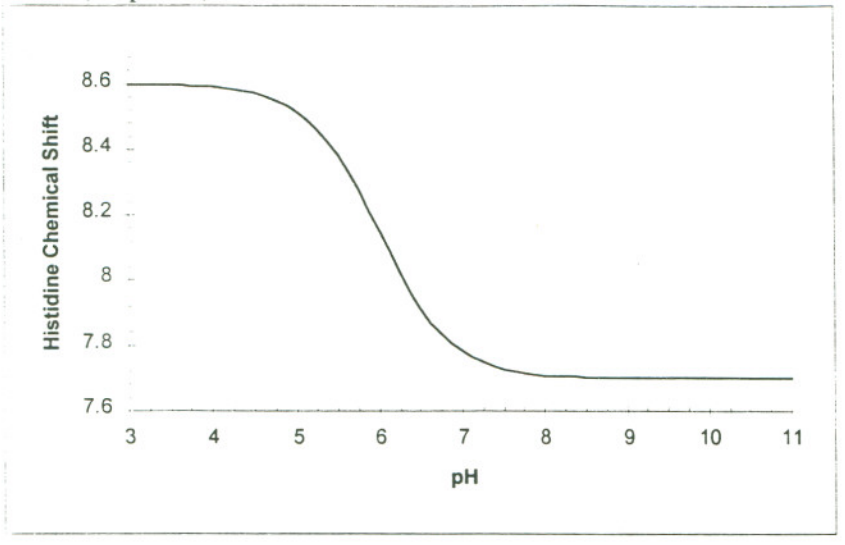
BIOCHEMISTRY CUMULATIVE EXAM

April 20, 1996

1. The NMR signal of histidine is well resolved from other peaks and there is a nice shift in peak position as the protonation state of the imidazole side chain changes. Because of this it is possible to easily titrate a protein sample and determine histidine pK_as. Shown below is such a titration for a particular histidine in the protein examase. By inspection of this graph, what is the pK_a of this histidine to the nearest 0.25 pH unit? (20 points)



2. The pK_a of the model compound, N-acetyl-histidine-N-methylamide, is near 7.0. The pK_a of side chain functional groups is often perturbed in proteins from the values one finds in simple model compounds because of interactions with other groups. In the crystal structure of examase the histidine that was titrated above has the amino group of a lysine near to it. Could this explain the difference in pK_a of this particular histidine compared to that of N-acetyl-histidine-N-methylamide? Explain briefly why or why not. (20 points)

3. Assume the "standard" pK_a of histidine is 7.0. In free energy terms, how much harder or easier is it to protonate the histidine in examase? (25 points)

4. You mutate an aspartic acid in examase to an asparagine. You determine the stability of the wild-type protein and the asparagine mutant at neutral pH and find that the mutant is less stable than the wild-type protein. Predict whether the pK_a of the aspartate in the protein is likely to be higher or lower than the pK_a of the side chain carboxyl of N-acetyl-aspartate-N-methylamide. Explain your answer briefly. (30 points)

Some equations that may (or may not) be of use:

For the reaction $HA \rightleftharpoons H^+ + A^-$

A) $K = [H^+][A^-]/[HA]$

B) $pH = pK_a + \log_{10}([A^-]/[HA])$

C) $pH = -\log_{10}[H^+]$

D) $\Delta G = -RT\ln K$ $\quad R = 1.98 \times 10^{-3}$ kcal/mol deg $\quad$ Assume T = 298°K

5. Which of the equations above, A, B, C, or D, is the Henderson-Hasselbalch equation? (5 points)